

Teaching the Transportation Problem with Spatial Data

Prasad Gavande	202401173
Shivanshu Singh	202401230
Omkar Fulkar	202402093
Parmar Manank	202402097
Saharsha Bartakke	202402126

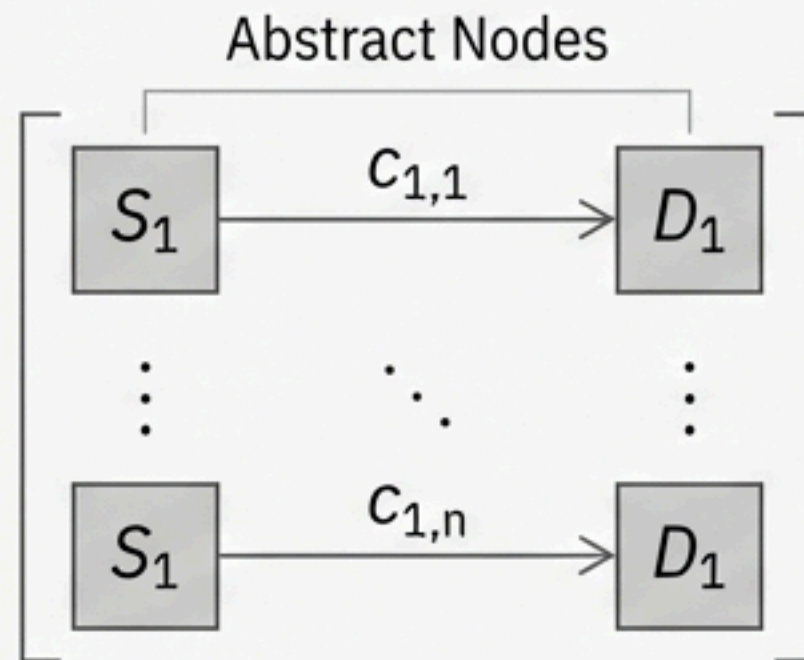
Synthesised from the work of Jyotirmoy Dalal (Indian Institute of Management Lucknow), published in *INFORMS Transactions on Education* (2022).



Bridging the gap between abstract textbook matrices and the messy reality of last-mile logistics.

The Pedagogical Gap: Why Abstract Matrices Fail to Engage

The Textbook Approach



- Standard Operations Research (OR) curricula rely on pre-given matrices.
- Focuses heavily on theoretical properties (integrality, network simplex).
- Ignores the complexity of data generation.
- Result: Disconnect between 'solving the model' and 'designing the network'.

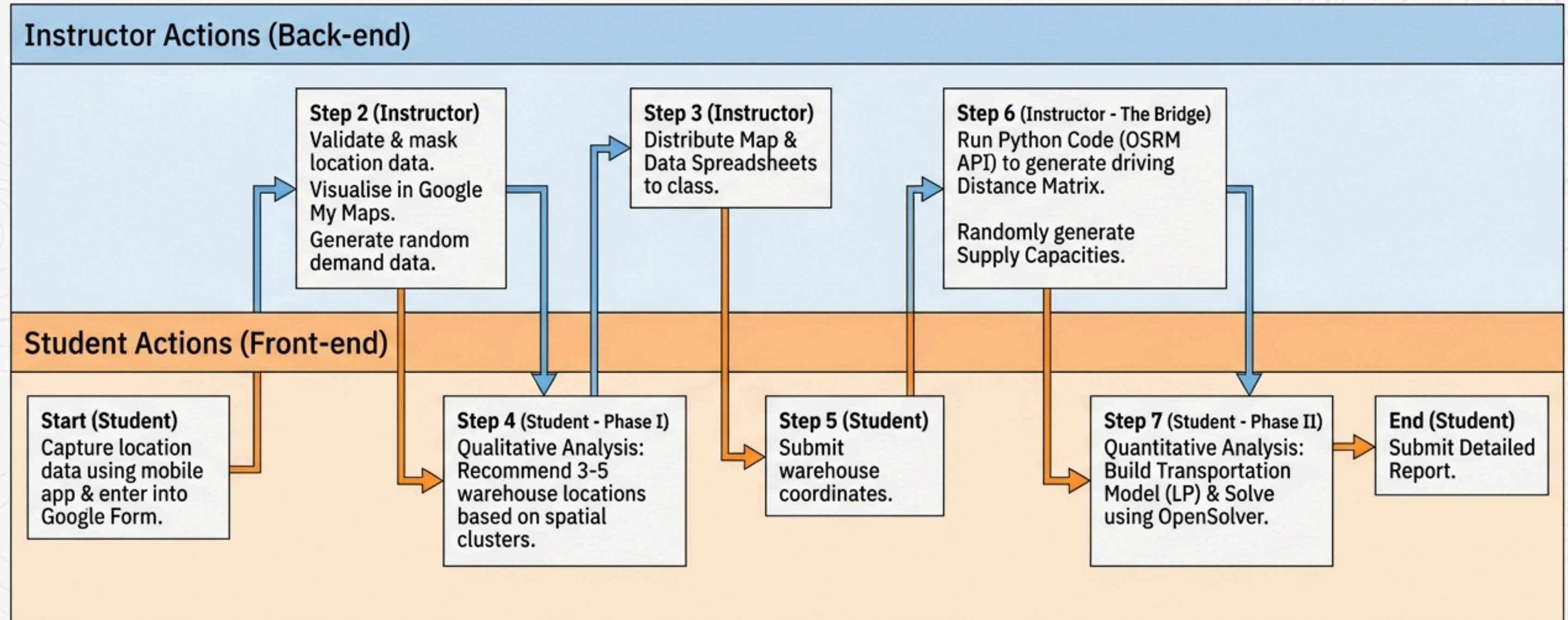
The Real-World Gap



- Post-COVID-19 logistics: Phenomenal growth in online market demand.
- Student Demand: Business students crave practical applicability over methodological nuances.
- The Solution: Students become the nodes. By utilising their own geographic locations, the class generates a realistic spatial dataset, fostering immediate emotional investment.

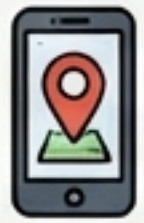
The Exercise Architecture: A Two-Phase Playbook

Timeline: Approx. 2 Weeks | Structure: Student-Instructor Handshake



The Toolkit: Open Source & Accessible

Student Requirements (Front-End Tools)



Smartphones: GPS Apps to capture Latitude/Longitude.

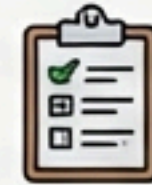


Google My Maps: For visualising demand clusters and spatial planning.



Spreadsheet Software: Excel or LibreOffice Calc.

Instructor Requirements (Back-End Tools)



Google Forms: For efficient data aggregation.



Python Script: *calcDM.py* (Provided) to automate distance calculations.



OSRM API: Open Source Routing Machine for real road network data.



OpenSolver: Excel add-in to bypass the standard 200-variable limit.



Mise en place

Step 1: Crowdsourcing the Network

Converting students into data points.

The Task: Students capture their current Latitude/Longitude using a GPS app and submit it via a Google Form.

Rich Data: Beyond location, students rank preferences for four product categories:

1. Electronics
2. Books
3. Apparel
4. Grocery

Why it Works: In a remote or hybrid class, students are dispersed. This creates a realistic, pan-national distribution of “customers”.

Field	Value
Enter Name (optional)	XXXXX
Enter State Name	Andhra Pradesh
Enter Pincode (digits only)	517502
Enter Latitude of your current location (up to 8 decimal point precision)	13.619065
Enter Longitude of your current location (up to 8 decimal point precision)	79.324446

Self-Generated Spatial Data

Rank	Electronics	Book	Apparel	Grocery
First Choice	X			
Second Choice			X	
Third Choice		X		
Fourth Choice				X

Data Prep: Validation, Masking, and Demand Logic

Instructor Actions (Back-end)



Privacy & Validation

Masking: Real coordinates are perturbed by adding a random fraction to entries.

Validation: Import raw CSV to Google My Maps to spot errors (e.g., missing decimals).



The Logic Engine

Converting text preferences to numeric demand.

```
IF(Choice="Books", 4,  
IF(Choice="Electronics", 3...))  
RANDBETWEEN(350, 450) if Rank=1
```



The Output

A clean dataset with 100+ Demand Points (DP1...DP101) ready for distribution.

Mise en place

Phase I: Visualising the Demand Landscape

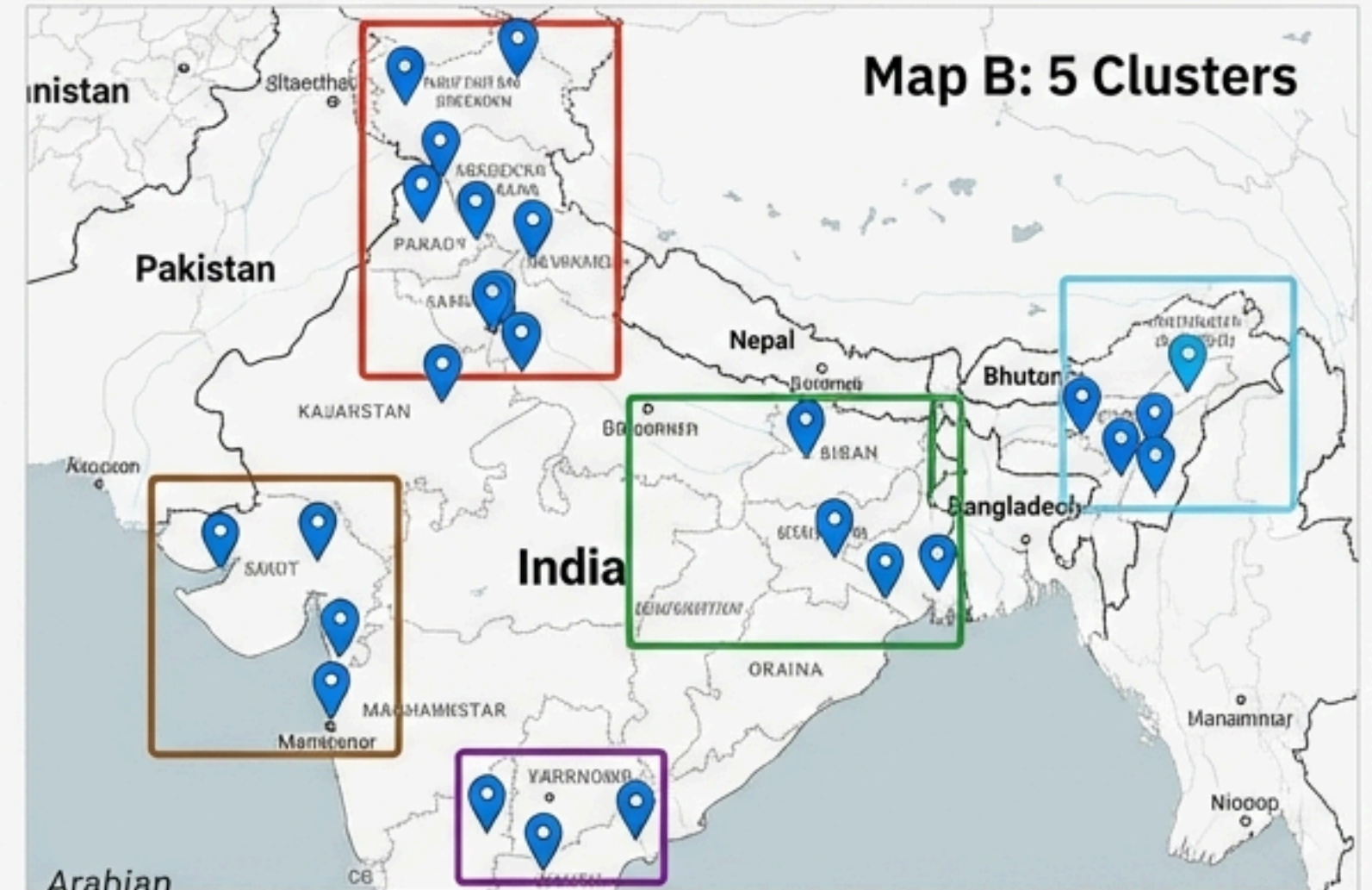
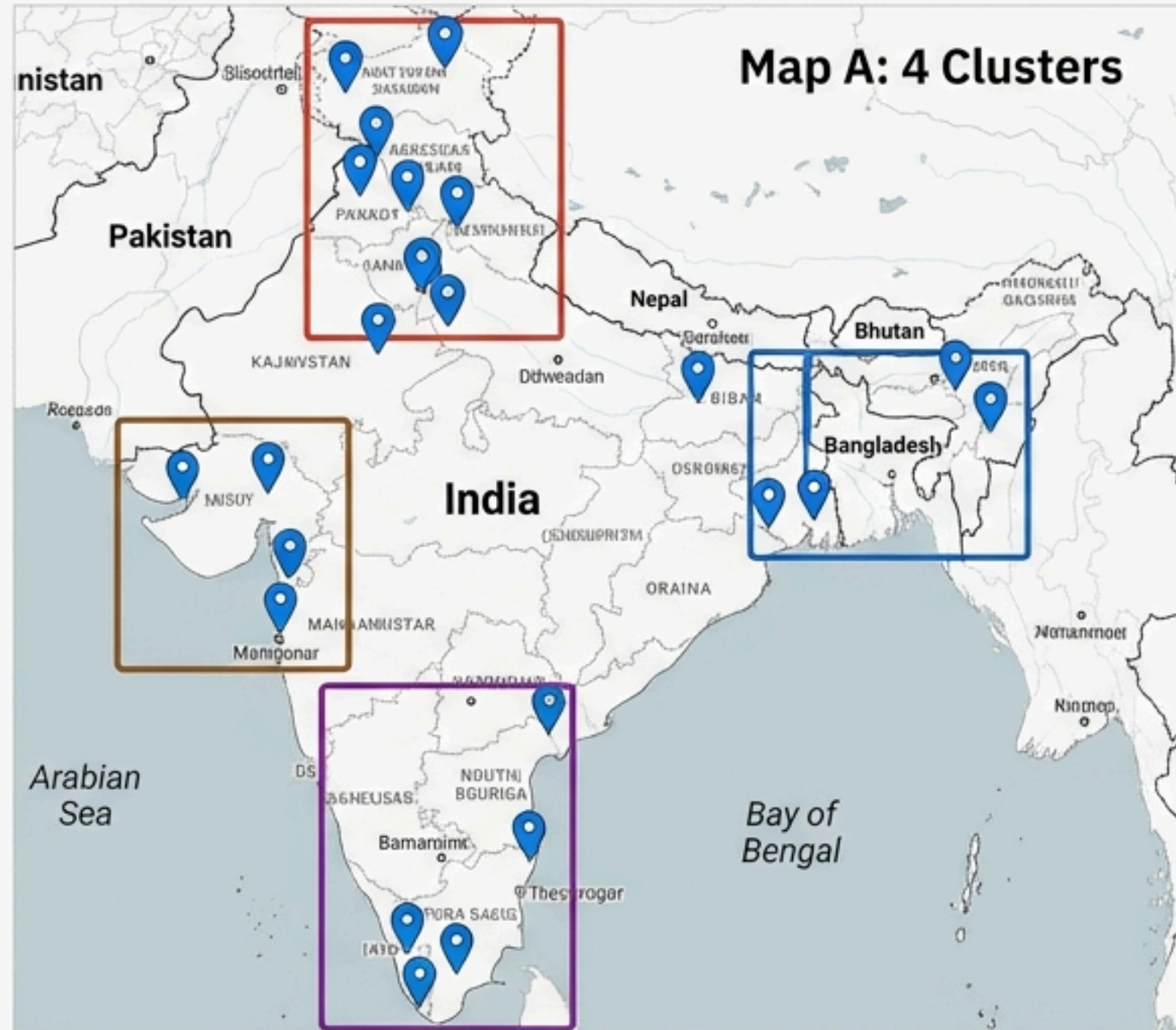


The Student View:
Students receive the dataset and the map visualisation.

The Task: 'You are seeing 101 customer locations. Identify the clusters.'

Pedagogical Insight:
Visualisation allows students to spot density patterns immediately—North, South, East, West, and Central hubs—turning a spreadsheet into a geography problem.

The Strategic Challenge: Locating the Warehouses



The Prompt:

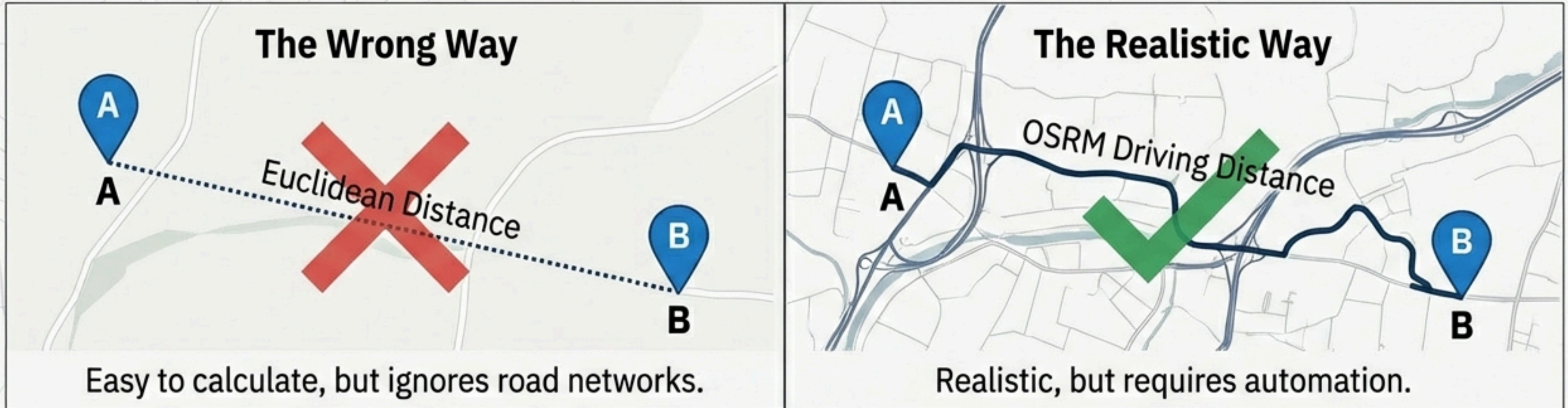
Recommend 3 to 5 warehouse locations to serve the network.

The Human Element:

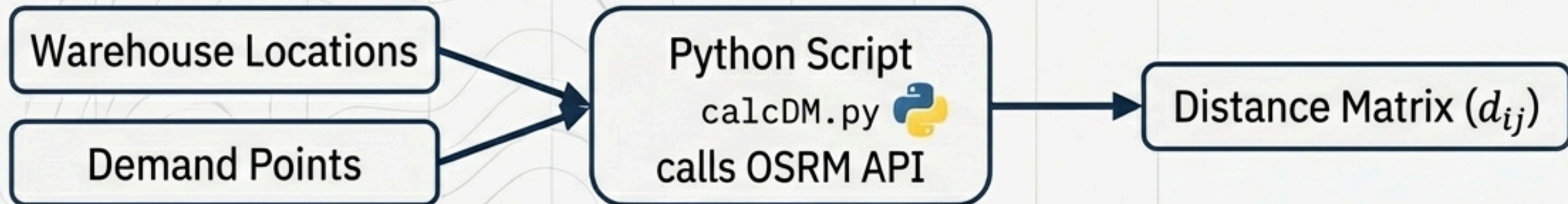
Students must move beyond math to consider:

- **Proximity:** Centroids of demand (Visual or K-means).
- **Infrastructure:** Access to highways/transport hubs.
- **Policy:** State tax structures and SEZs.

The Bridge: From Euclidean to Real-World Driving Distances

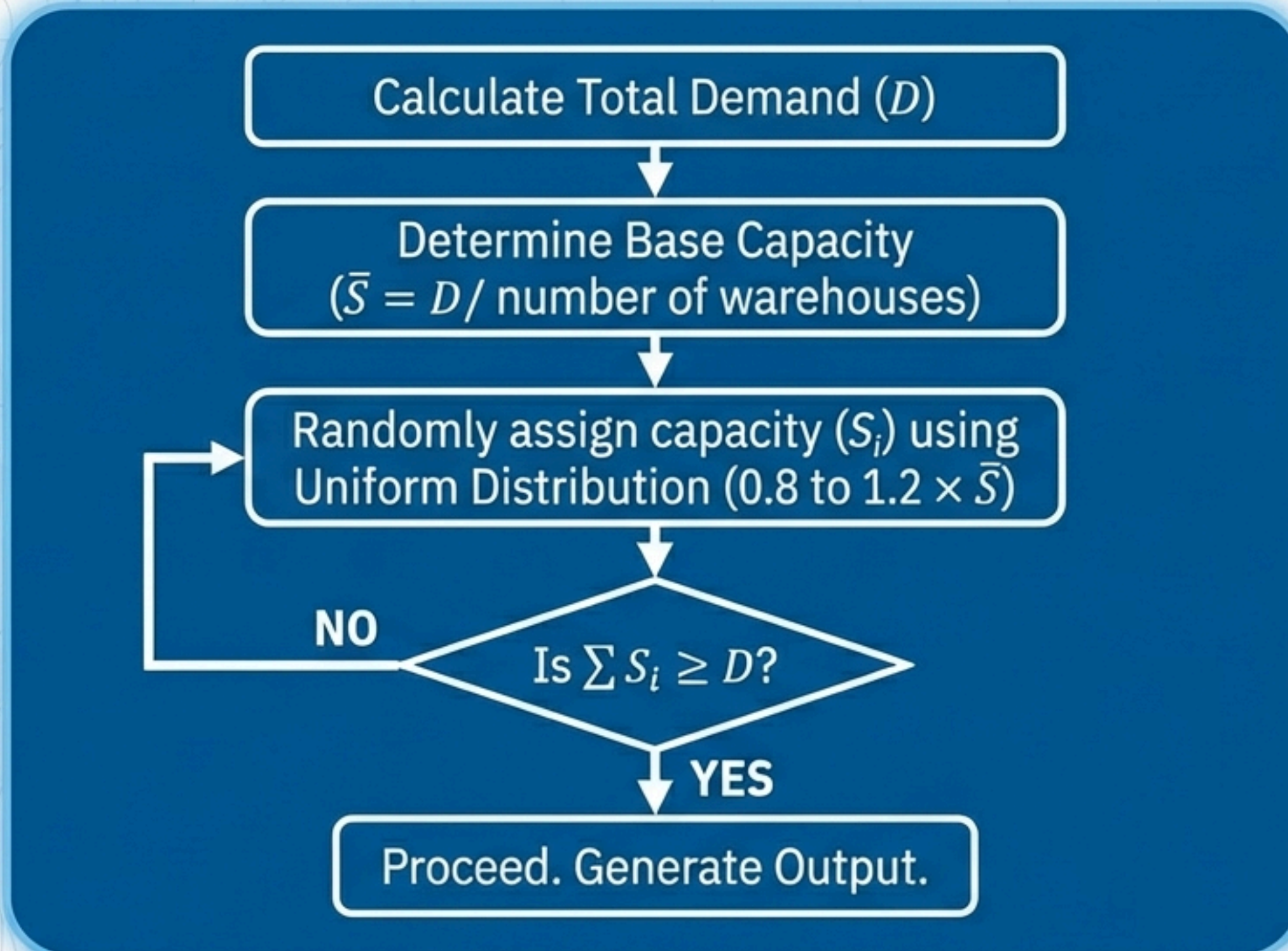


The Automation Solution (Instructor Python Script)



This automation allows the exercise to remain realistic without becoming tedious data entry work.

Ensuring Feasibility: The Capacity Algorithm



In a “No Shortage” model, Total Supply must be $\geq$ Total Demand.

This instructor-led step guarantees the student student models will be solvable.

Phase II: The Mathematical Formulation

Linear Programming Model

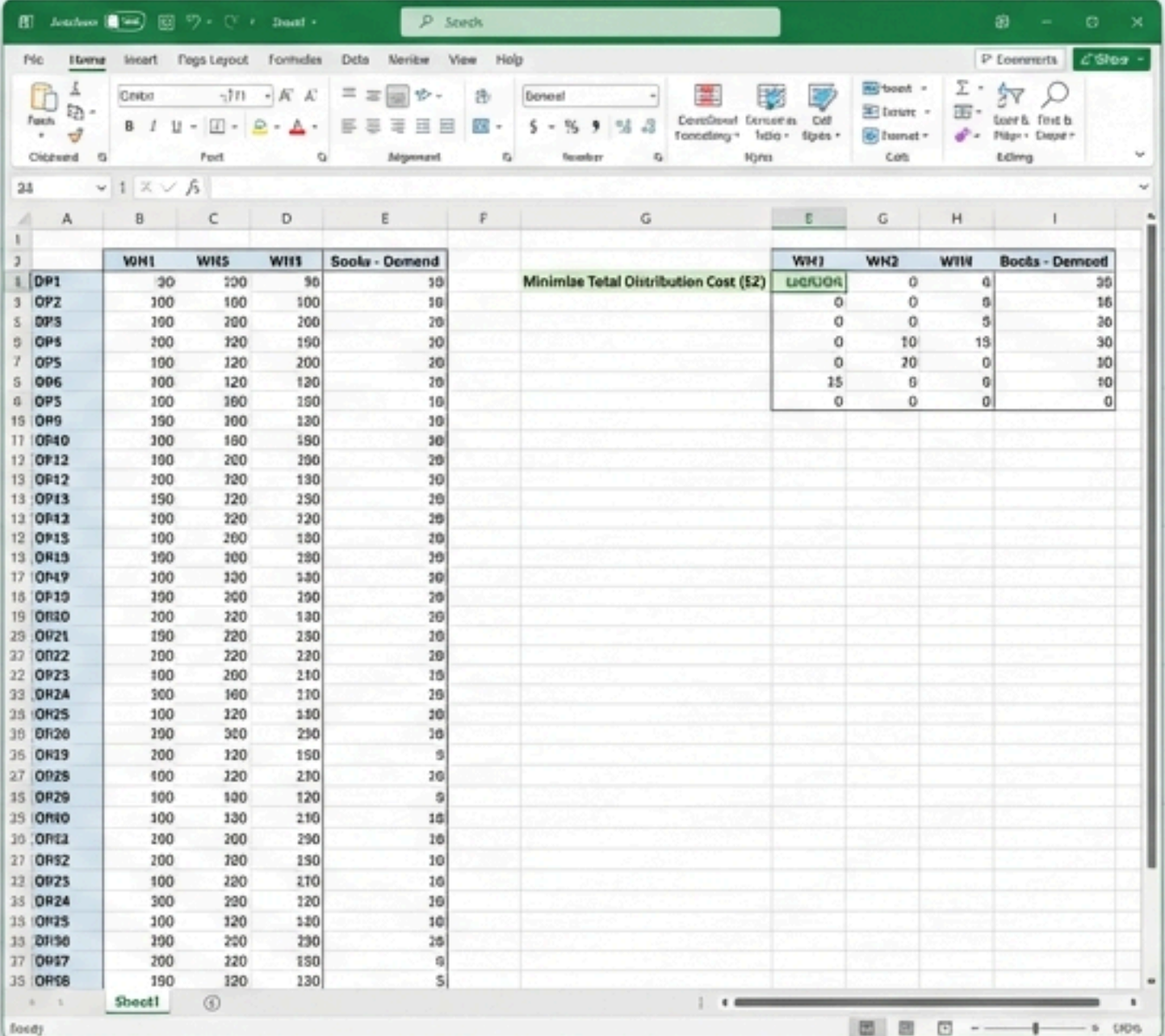
Objective: Minimize Total Distribution Cost (Z)

$$Z = \sum \sum c_{ij} x_{ij}$$

(where c_{ij} is cost based on driving distance)

Constraints:

1. **Supply:** $\sum x_{ij} \leq S_i$ (Outflow $\leq$ Capacity)
2. **Demand:** $\sum x_{ij} \geq D_j$ (Inflow $\geq$ Demand)
3. **Non-negativity:** $x_{ij} \geq 0$



	WH1	WH2	WH3	Books - Demand
DP1	30	200	90	10
DP2	100	100	100	10
DP3	100	200	200	20
DP4	200	120	190	20
DP5	100	120	200	20
DP6	100	120	130	20
DP7	100	100	190	10
DP8	190	100	130	10
DP9	100	100	190	20
DP10	100	200	190	20
DP11	100	200	130	20
DP12	190	120	250	20
DP13	100	220	220	20
DP14	100	200	190	20
DP15	190	100	190	20
DP16	100	100	190	20
DP17	100	200	290	20
DP18	100	200	220	20
DP19	100	200	210	10
DP20	100	100	110	10
DP21	100	120	190	20
DP22	100	100	290	10
DP23	100	200	210	10
DP24	100	200	190	10
DP25	100	200	210	10
DP26	100	200	210	10
DP27	100	200	210	10
DP28	100	200	210	10
DP29	100	200	210	10
DP30	100	200	210	10
DP31	100	200	210	10
DP32	100	200	210	10
DP33	100	200	210	10
DP34	100	200	210	10
DP35	100	200	210	10
DP36	100	200	210	10
DP37	100	200	210	10
DP38	100	200	210	10
DP39	100	200	210	10
DP40	100	200	210	10

Scaling Up: Why We Need OpenSolver



Standard Excel Solver

Limit: 200 Decision Variables.

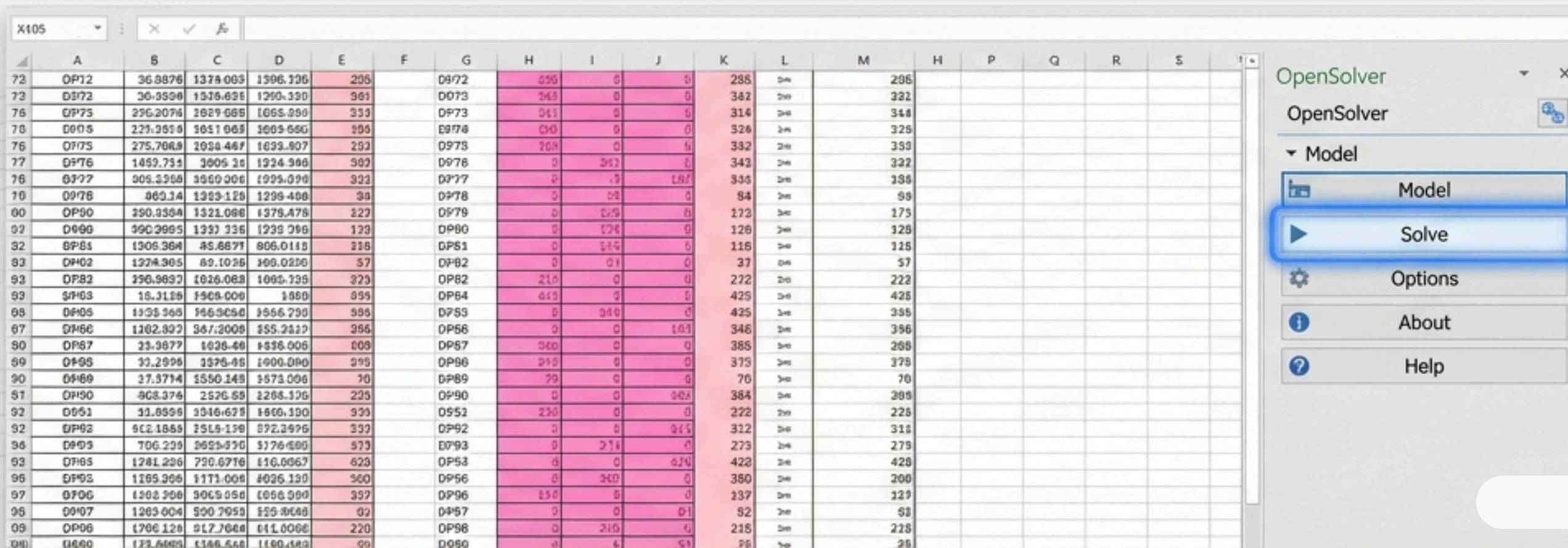
Insufficient for 100 demand points x 5 warehouses (500 variables).



OpenSolver

Limit: No artificial limits (Linear/Integer).

Why: Essential for solving real-world scale logistics problems.

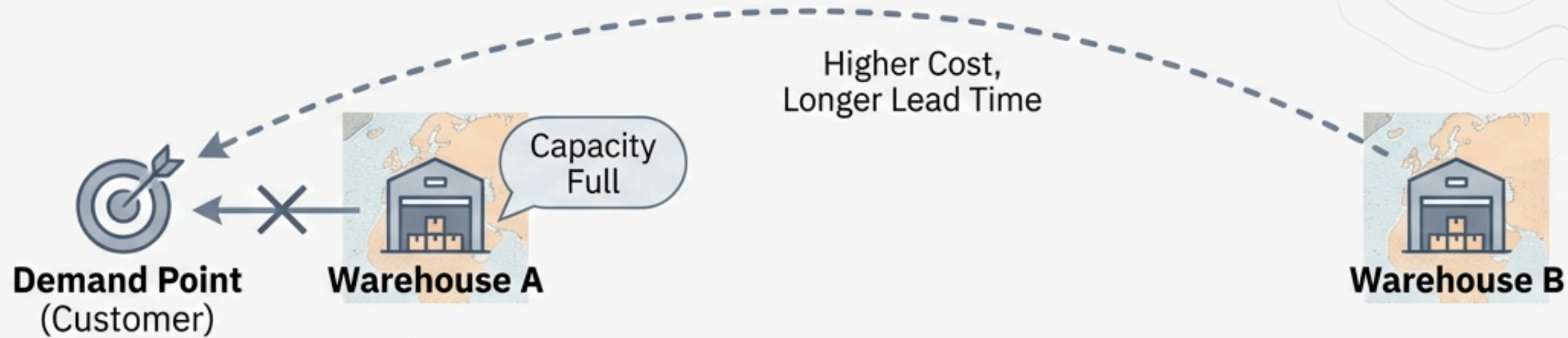


The screenshot shows an Excel spreadsheet with columns A through S and rows 72 through 99. The cells contain numerical data, with some cells highlighted in pink. The OpenSolver interface is overlaid on the right side of the spreadsheet. The interface includes a title bar 'OpenSolver' and a menu bar with 'OpenSolver' and a gear icon. Below the menu bar is a 'Model' section with a 'Model' button and a 'Solve' button. The 'Solve' button is highlighted with a blue border. Below the 'Solve' button are buttons for 'Options', 'About', and 'Help'.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
72	DP12	36.8876	1374.063	1506.136	206		DP72	0.00	0	0	288	0.00	206						
73	DP72	36.8876	1374.063	1506.136	361		DP73	0.00	0	0	362	0.00	322						
76	DP73	276.2076	1629.666	1665.290	333		DP73	0.00	0	0	314	0.00	344						
78	DP05	223.3518	1617.063	1603.660	103		DP74	0.00	0	0	326	0.00	326						
76	DP75	275.7068	1634.467	1633.807	293		DP75	0.00	0	0	332	0.00	333						
77	DP76	1453.733	1605.21	1524.948	303		DP76	0	0.00	0	343	0.00	322						
76	DP77	905.3366	1600.006	1595.090	333		DP77	0	0	1.87	333	0.00	338						
78	DP78	860.14	1583.126	1596.488	38		DP78	0	0.00	0	84	0.00	98						
80	DP90	290.9384	1521.086	1578.478	223		DP79	0	0.00	0	173	0.00	173						
82	DP80	390.3665	1337.336	1233.086	133		DP80	0	0.00	0	126	0.00	126						
82	DP81	1306.364	45.6677	806.0113	218		DP81	0	0.00	0	116	0.00	116						
83	DP82	1324.365	89.1026	108.0250	57		DP82	0	0.00	0	37	0.00	57						
83	DP82	290.9637	1026.063	1086.735	323		DP82	21.0	0	0	272	0.00	222						
83	DP83	18.3186	1908.000	1880	893		DP84	613	0	0	425	0.00	425						
88	DP85	1335.366	1463.058	1566.730	593		DP85	0	0.00	0	425	0.00	355						
87	DP86	1102.927	1671.2008	155.2117	386		DP86	0	0	1.03	346	0.00	396						
90	DP87	23.3677	1626.46	1536.006	603		DP87	0.00	0	0	385	0.00	208						
89	DP88	33.2995	1375.46	1400.000	293		DP88	0.00	0	0	373	0.00	373						
90	DP89	27.3714	1550.145	1673.006	70		DP89	0	0	0	70	0.00	70						
91	DP90	408.376	2626.69	1265.136	293		DP90	0	0	0.00	384	0.00	289						
92	DP91	31.8996	1316.628	1400.130	333		DP91	230	0	0	222	0.00	226						
92	DP92	612.1853	1515.130	872.2976	333		DP92	0	0	0.00	312	0.00	312						
96	DP93	706.233	1621.370	1776.600	573		DP93	0	2.14	0	273	0.00	273						
93	DP94	1241.226	720.6776	110.0067	623		DP94	0	0	0.00	428	0.00	428						
96	DP95	1185.306	1171.006	4026.130	560		DP95	0	0.00	0	380	0.00	200						
97	DP96	1185.306	1065.056	1066.000	337		DP96	136	0	0	237	0.00	123						
98	DP97	1263.604	900.7053	155.8068	0		DP97	0	0	0.00	92	0.00	63						
98	DP98	1706.126	917.7044	611.0066	220		DP98	0	0.00	0	218	0.00	218						
99	DP99	173.6689	1146.646	1180.183	0		DP99	0	0	0.00	26	0.00	26						

The "Aha!" Moment: Cost vs. Responsiveness

The Insight: Students realise the “closest” warehouse is not always the supplier.



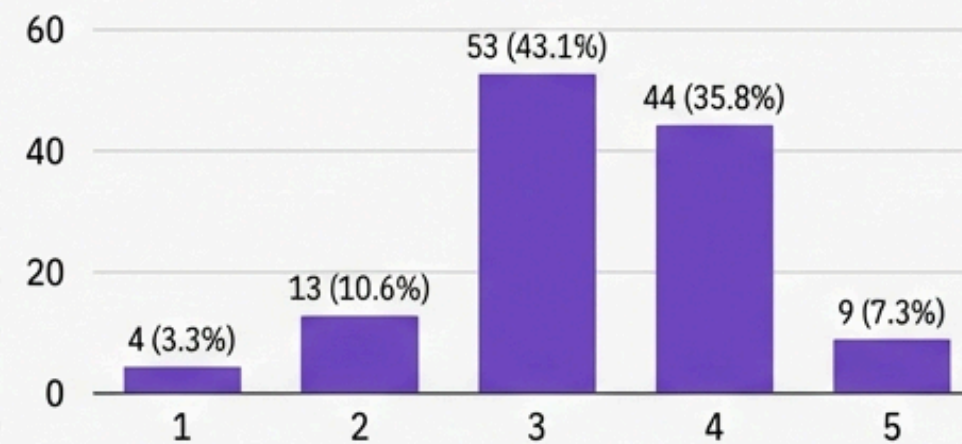
Summary Results: Cost Variance Based on Warehouses

Group	Objective Function Value (Total Cost)	# of Warehouses
1st Group 4	INR 14,889,335.00	3
2nd Group 8	INR 12,111,353.00	4
3rd Group 12	INR 12,737,260.00	4
4th Group 16	INR 12,256,156.00	4

Thank You

Student Feedback & Pedagogical Wins

Prior Quantitative Exposure



Stream

Engineering	66.7%
Commerce	9.8%
Economics	6.5%
Management	6.5%
Agriculture	4.8%
Science	4.1%
Biotechnology	1.6%
Total	100.0%

The assignment simulated real life problems... [and] helped in understanding perspectives of team members.

We could see the real locations... realising how important and effective OR methods can be.

Key Takeaways: Ownership of data, Integrated Strategic/Tactical decision making, and Visualisation as an analytical tool.